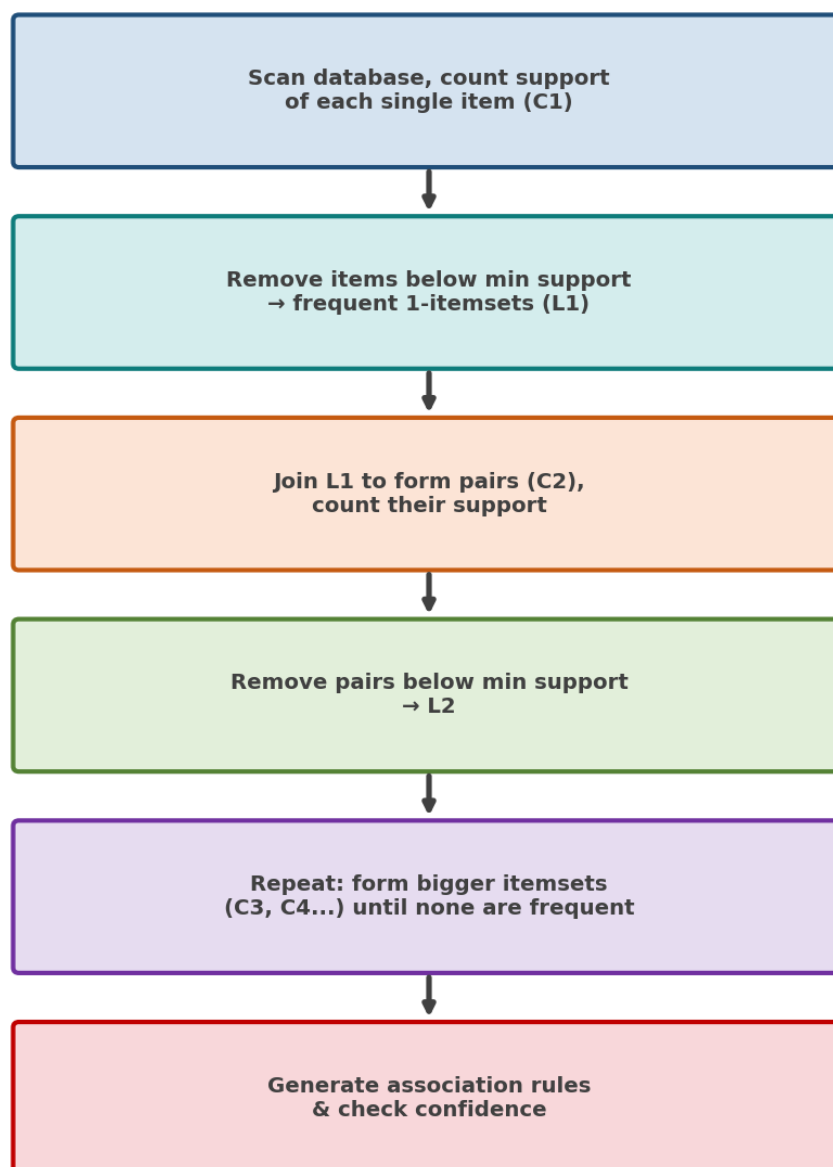


1. What is Association Rule Mining? Explain the terms: Support, Confidence, and Lift. ★



Apriori Algorithm — Flow



Association Rule Mining is a data mining technique used to **find relationships or patterns between items** in a large dataset. It tells us which items are often bought or used together. It is widely used in **Market Basket Analysis** (e.g., "people who buy bread also buy butter").

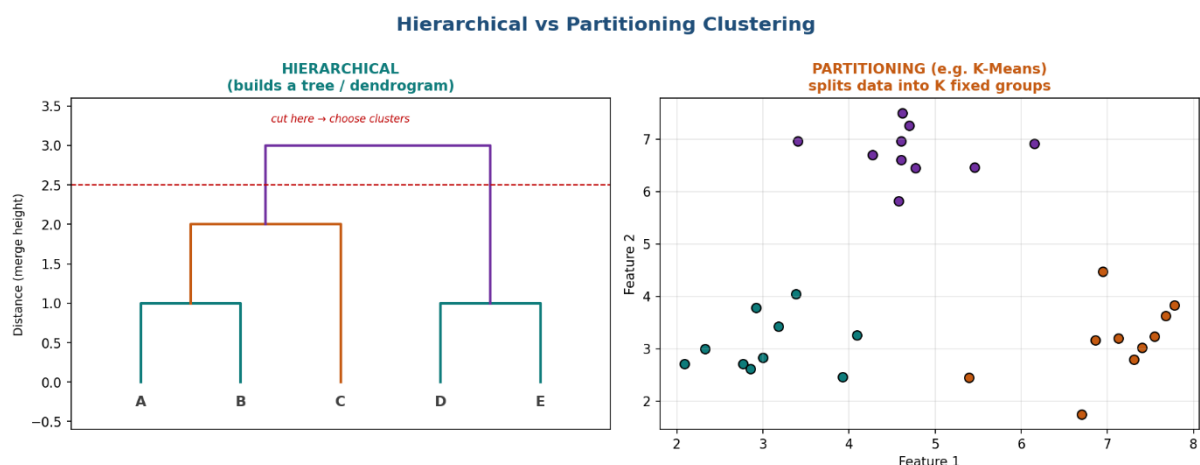
A rule is written as $A \rightarrow B$, meaning "if A is bought, then B is likely bought too."

Three key terms:

1. **Support** — How often the items appear together in all transactions. It shows how popular an itemset is.
 - Formula: $\text{Support}(A) = (\text{Transactions containing } A) \div (\text{Total transactions})$
 - *Example:* If bread and butter appear together in 200 out of 1,000 bills, support = $200/1000 = 0.20$ (20%).
2. **Confidence** — How often B is bought **when** A is bought. It shows how reliable the rule is.
 - Formula: $\text{Confidence}(A \rightarrow B) = \text{Support}(A \text{ and } B) \div \text{Support}(A)$
 - *Example:* If 80% of people who buy bread also buy butter, confidence = 80%.
3. **Lift** — How much **more likely** B is bought when A is bought, compared to B alone. It shows the strength of the link.
 - Formula: $\text{Lift}(A \rightarrow B) = \text{Confidence}(A \rightarrow B) \div \text{Support}(B)$
 - **Lift** > 1 → positive link (good rule), = 1 → no link, < 1 → negative link.

Conclusion: Support shows popularity, confidence shows reliability, and lift shows the real strength of the relationship.

2. What is the difference between Hierarchical Clustering and Partitioning method? ★ ★



Both are **clustering** methods (grouping similar data), but they work differently.

Hierarchical Clustering builds a **tree of clusters** (called a dendrogram), step by step.

- **Agglomerative (bottom-up):** start with each point as its own cluster, then keep merging the closest ones.
- **Divisive (top-down):** start with one big cluster and keep splitting it.
- You do **not** need to decide the number of clusters in advance — you "cut" the tree at any level.

Partitioning Method divides the data into a **fixed number (K) of clusters** directly (e.g., K-Means).

- You **must decide K** in advance.
- It keeps adjusting points until clusters are stable.

Comparison:

Point	Hierarchical	Partitioning (K-Means)
Output	Tree (dendrogram)	K flat groups
Number of clusters	Not needed in advance	Must be given (K)
Speed	Slower (for big data)	Faster
Re-grouping	Cannot undo a merge	Points can move between clusters
Best for	Small data, nested groups	Large data

Conclusion: Hierarchical gives a flexible tree view, while partitioning quickly splits data into a set number of groups.

3. Explain Bayes Theorem in detail. ★ ★

Bayes Theorem is a formula that finds the **probability of an event based on prior knowledge** of related conditions. It updates a belief when new evidence is available.

Formula: $P(A|B) = \{P(B|A) * P(A)\} / P(B)$

In simple words:

- $P(A|B)$ = chance of A happening, given that B has happened (what we want).
- $P(B|A)$ = chance of B happening, given A.
- $P(A)$ = chance of A alone (prior).
- $P(B)$ = chance of B alone.

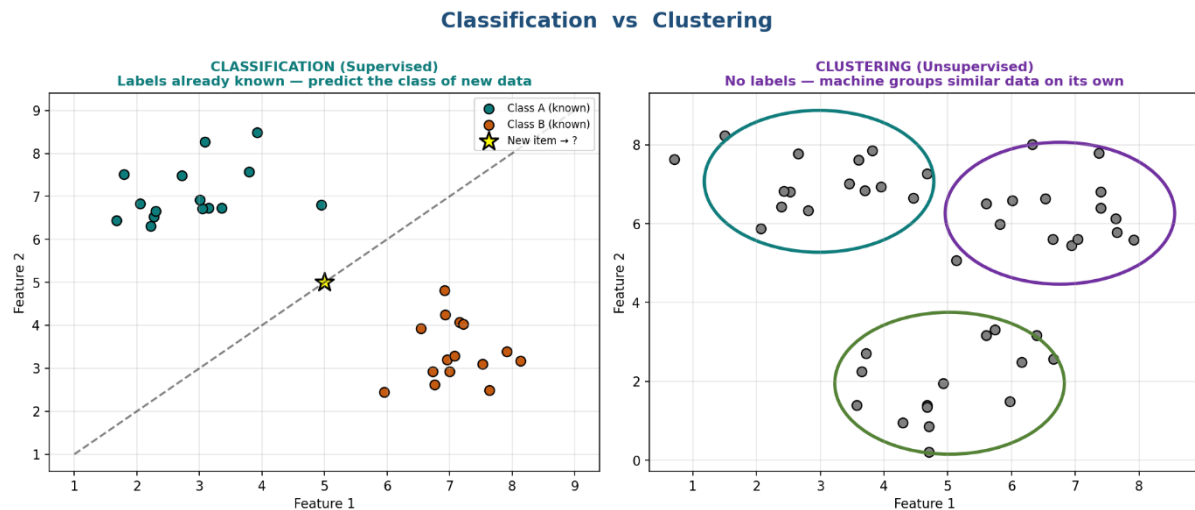
Example: A test detects a disease.

- 1% of people have the disease $\rightarrow P(\text{Disease}) = 0.01$.
- If the test is positive, Bayes Theorem combines the disease rate and the test accuracy to find the **real chance the person actually has the disease**.

Use in BI / Data Mining: It is the base of the **Naïve Bayes classifier**, used for spam filtering, sentiment analysis, and medical diagnosis.

Conclusion: Bayes Theorem helps make better predictions by combining old knowledge with new evidence.

4. Difference between Classification and Clustering. ★★★★★



Both are data mining techniques, but they work very differently.

Classification is a **supervised** technique. The data already has **labels (known classes)**, and the model learns from them to **predict the class of new data**.

- *Example:* From past emails labeled "Spam" or "Not Spam", the model learns and then classifies a new email.

Clustering is an **unsupervised** technique. The data has **no labels**, and the machine **groups similar data on its own**.

- *Example:* Grouping customers by buying behaviour without knowing the groups in advance.

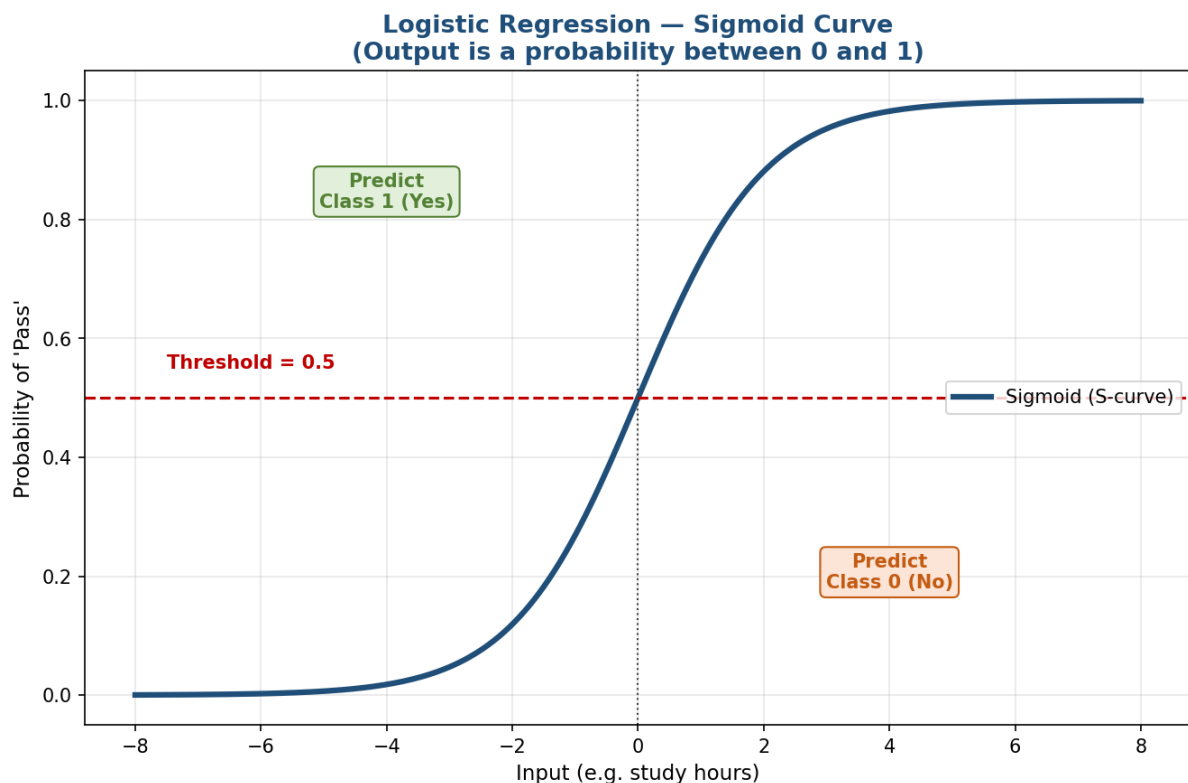
Comparison:

Point	Classification	Clustering
Type	Supervised (labels known)	Unsupervised (no labels)
Goal	Predict the class of new data	Group similar data
Training data	Labeled	Unlabeled
Output	Predefined classes	Natural groups (clusters)
Example	Spam vs Not Spam	Customer segments
Algorithms	Decision Tree, Naïve Bayes	K-Means, Hierarchical

Simple idea: In classification, the groups are **already known** and we sort new data into them. In clustering, the groups are **discovered** by the machine.

Conclusion: Classification needs labeled data to predict, while clustering finds hidden groups in unlabeled data.

5. Explain Logistic Regression with example. ★ ★ ★ ★



Logistic Regression is a **classification** technique used to predict a **category (Yes/No, 0/1)**, not a number. Even though it has "regression" in its name, it is used for classification.

How it works:

- It takes inputs and calculates a **probability between 0 and 1** using a special **S-shaped curve** called the **sigmoid function**.
- A **threshold (usually 0.5)** decides the class:
 - If probability $\geq 0.5 \rightarrow$ Class 1 (Yes)
 - If probability $< 0.5 \rightarrow$ Class 0 (No)

Sigmoid formula (simple): $P = \frac{1}{1 + e^{-z}}$ where z is the input value.

Example: Predicting if a student will **pass** based on study hours.

- A student who studies many hours $\rightarrow$ probability = 0.9 $\rightarrow$ **Pass (Class 1)**.
- A student who studies very few hours $\rightarrow$ probability = 0.2 $\rightarrow$ **Fail (Class 0)**.

- The S-curve smoothly turns study hours into a pass/fail decision.

Where used: Spam detection, disease prediction (sick/healthy), pass/fail, customer will buy/not buy.

Conclusion: Logistic regression gives a probability and uses a threshold to classify data into one of two groups.

6. What is Logistic Regression? Discuss the types of Logistic Regression. ★ ★

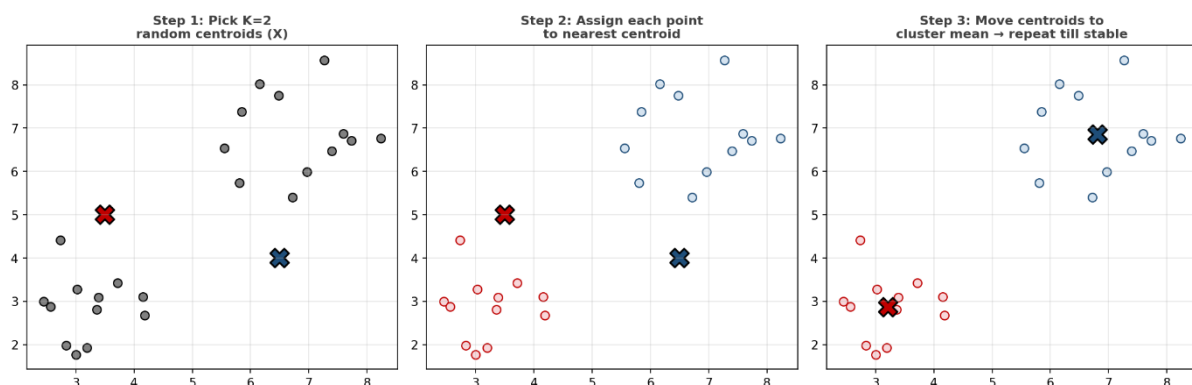
Logistic Regression is a classification method that predicts categories using probability. Based on the **number of output categories**, it has three types:

1. **Binary (Binomial) Logistic Regression** — The output has only **two categories**.
 - *Example:* Pass/Fail, Spam/Not Spam, Yes/No.
2. **Multinomial Logistic Regression** — The output has **three or more categories** that have **no order**.
 - *Example:* Predicting a fruit as Apple, Banana, or Mango.
3. **Ordinal Logistic Regression** — The output has **three or more categories** that have a **natural order (ranking)**.
 - *Example:* Movie rating as Poor → Average → Good → Excellent, or Low/Medium/High.

Conclusion: Binary is for two options, multinomial is for many unordered options, and ordinal is for many ordered options.

7. What is K-Means Clustering? Explain the step-by-step working of the K-Means algorithm with example. ★ ★

K-Means Clustering — Step by Step (K = 2)



K-Means is an **unsupervised** clustering method that divides data into **K groups (clusters)**, where each group has similar data points.

Step-by-step working:

1. **Choose K** — decide how many clusters you want (e.g., $K = 2$).
2. **Pick K centroids** — select K random points as the starting cluster centers.
3. **Assign points** — put each data point into the cluster of its **nearest centroid** (using distance).
4. **Update centroids** — move each centroid to the **average (mean)** position of its cluster's points.
5. **Repeat** steps 3 and 4 until the centroids **stop moving** (clusters become stable).

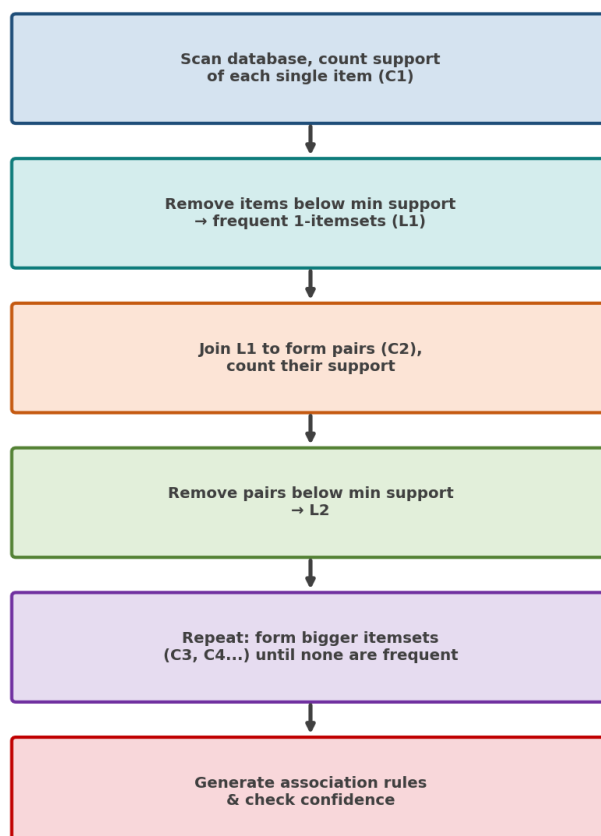
Example: Grouping customers by age and income into 2 clusters — the algorithm keeps adjusting until it finds two stable groups (e.g., "young low-income" and "older high-income").

Conclusion: K-Means keeps assigning points and updating centers until it forms K clear, stable groups. It is simple and fast, but you must choose K in advance.

8. What is the Apriori Algorithm? Discuss the applications of this algorithm with example.

★ ★

Apriori Algorithm — Flow



The **Apriori Algorithm** finds items that are **frequently bought or used together** and creates association rules. It works on the idea that *if an itemset is frequent, its subsets are also frequent*.

Applications:

1. **Market Basket Analysis** — finding products bought together (e.g., bread + butter) to plan store layout and combo offers.
2. **Recommendation systems** — "Customers who bought this also bought..." on shopping websites.
3. **Cross-selling** — suggesting related products to increase sales.
4. **Medical diagnosis** — finding symptoms that often occur together.
5. **Web usage mining** — finding pages users visit together.
6. **Fraud detection** — spotting unusual combinations of transactions.
7. **Inventory management** — keeping related items in stock together.

Example: A supermarket finds that "milk → bread" has high support and confidence, so it places them near each other and offers a combo discount.

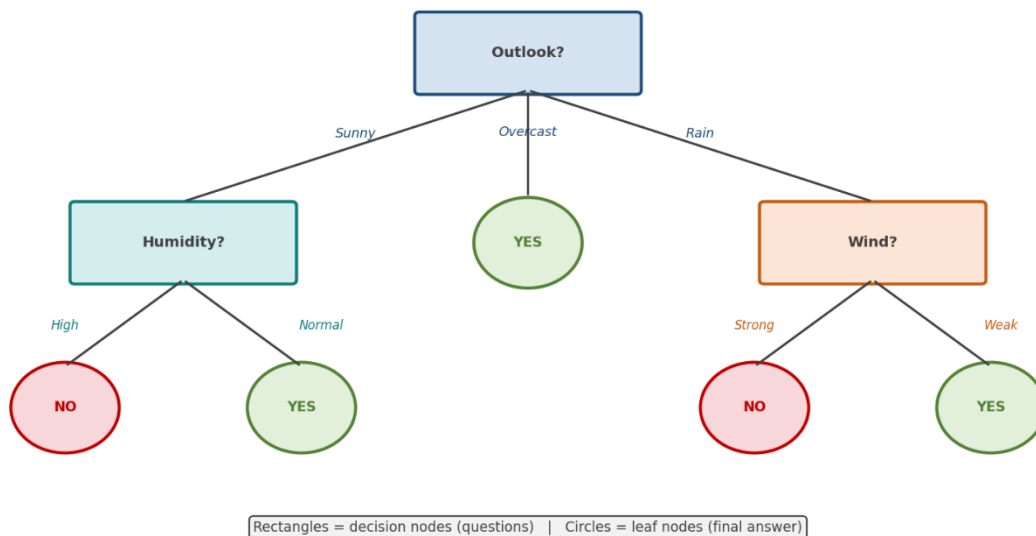
Conclusion: Apriori is mainly used wherever we need to find "**what goes with what**" to boost sales and improve decisions.

8. Define: Frequent Item Set, Minimum Support Count, Hierarchical Clustering, Regression. ★ ★

1. **Frequent Item Set** — A group of items that **appears together often** in the data, i.e., whose support is **equal to or above the minimum support**.
 1. *Example:* If {bread, butter} appears in many bills, it is a frequent itemset.
2. **Minimum Support Count** — The **smallest number of times** an itemset must appear to be called "frequent." Itemsets below this count are ignored.
 1. *Example:* If min support count = 2, an itemset must appear in at least 2 transactions.
3. **Hierarchical Clustering** — A clustering method that builds a **tree of clusters (dendrogram)** by repeatedly merging or splitting groups. No need to fix the number of clusters in advance.
4. **Regression** — A technique used to **predict a continuous numeric value** based on input data, by finding the relationship between variables.
 1. *Example:* Predicting house **price** from its size and location.

9. What is a Decision Tree? Explain with a case study. ★ ★

Decision Tree — Example: Should we Play Tennis?



Decision Tree is a **classification** method shaped like a tree, where data is split step by step based on questions until a final decision is reached. It is easy to understand because it works like human decision-making.

Parts of a decision tree:

- **Root Node** — the first/top question (best attribute to split on).
- **Internal Nodes** — further questions (decision points).
- **Branches** — the answers/outcomes of a question.
- **Leaf Nodes** — the final decision/class.

Case Study — "Should we play tennis?" The tree asks questions in order:

1. Outlook?

- If **Overcast** → **Play = Yes**.
- If **Sunny** → ask **Humidity?** → High = No, Normal = Yes.
- If **Rain** → ask **Wind?** → Strong = No, Weak = Yes.

So for a Sunny + Normal-humidity day, the tree's final answer is **Yes, play tennis**.

Advantages: Easy to understand, no complex math, works for both categories and numbers.

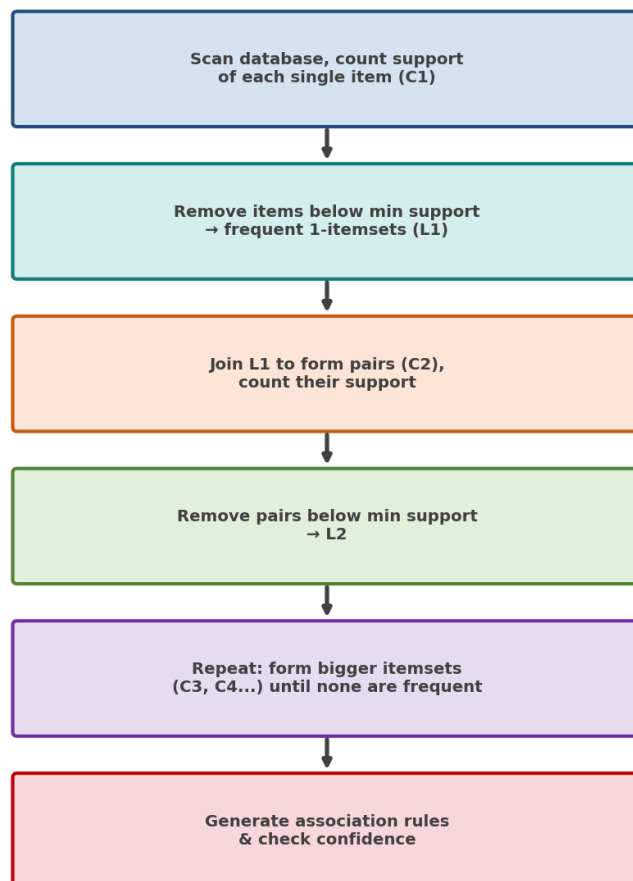
Disadvantage: Can become too complex (overfitting) if the tree grows too large.

Conclusion: A decision tree breaks a big decision into simple yes/no style questions, making classification clear and visual.

10. What are Association Rules? How to evaluate them using Support and Confidence?

Explain with example. ★ ★

Apriori Algorithm — Flow



Association Rules show **relationships between items** that often occur together, written as $A \rightarrow B$ ("if A, then B"). They are mainly used in market basket analysis.

We evaluate (judge) a rule using two measures:

1. **Support** — how often the items appear together in all transactions (popularity).
 - $\text{Support}(A \rightarrow B) = (\text{Transactions with both A and B}) \div (\text{Total transactions})$
 - *Example:* {milk, bread} in 300 of 1,000 bills → support = 0.30.
2. **Confidence** — how often B is bought when A is bought (reliability).
 - $\text{Confidence}(A \rightarrow B) = \text{Support}(A \text{ and } B) \div \text{Support}(A)$
 - *Example:* If 75% of milk buyers also buy bread → confidence = 75%.

How they are used together: We set a **minimum support** and **minimum confidence**. Only rules that cross **both** limits are kept as **strong rules**.

- *Example:* Rule "Milk $\rightarrow$ Bread" with support 30% and confidence 75% is strong, so the store places them together.

Conclusion: Support filters out rare item combinations, and confidence keeps only reliable rules — together they give useful, trustworthy patterns.

11. State different formulae for Evaluation of Classification models.

Confusion Matrix (Classification Evaluation)

Predicted $\rightarrow$		Positive	Negative
Actual	Positive	TP True Positive	FN False Negative
	Negative	FP False Positive	TN True Negative

Accuracy = $(TP+TN)/\text{Total}$

Recall = $TP/(TP+FN)$

Precision = $TP/(TP+FP)$

F1 = $2 \cdot (P \cdot R)/(P+R)$

To check how well a classification model works, we use a **Confusion Matrix** and formulas based on it.

Confusion Matrix terms:

- **TP (True Positive)** — correctly predicted Yes.
- **TN (True Negative)** — correctly predicted No.
- **FP (False Positive)** — wrongly predicted Yes.
- **FN (False Negative)** — wrongly predicted No.

Evaluation formulae:

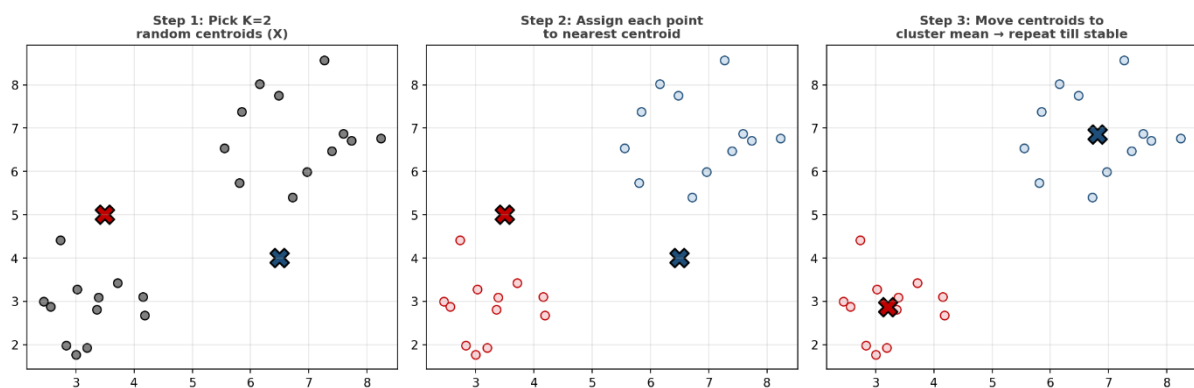
Measure	Formula	Meaning
Accuracy	$(TP + TN) \div \text{Total}$	Overall correct predictions
Precision	$TP \div (TP + FP)$	Of predicted Yes, how many were right
Recall (Sensitivity)	$TP \div (TP + FN)$	Of actual Yes, how many were found
F1-Score	$2 \times (\text{Precision} \times \text{Recall}) \div (\text{Precision} + \text{Recall})$	Balance of precision and recall
Error Rate	$(FP + FN) \div \text{Total}$	Overall wrong predictions

Example: If a spam filter has high precision, almost everything it marks as spam really is spam. If it has high recall, it catches almost all spam emails.

Conclusion: No single number is enough — accuracy, precision, recall, and F1 together give a complete picture of model performance.

12. Find clusters for the given website visitor age dataset using K-Means with K = 2: {16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66}.

K-Means Clustering — Step by Step (K = 2)



Data: 16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66 (19 values)

Step 1: Choose K = 2. Take starting centroids C1 = 16, C2 = 66.

Step 2 (Iteration 1): Assign each value to the nearer centroid.

- Cluster 1 (near 16): 16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41 → mean = **23.5**
- Cluster 2 (near 66): 42, 43, 44, 45, 61, 62, 66 → mean = **51.86**

Step 3 (Iteration 2): Re-assign using new centroids (23.5 and 51.86).

- Cluster 1: 16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41 → mean = **23.5** (no change)

- Cluster 2: 42, 43, 44, 45, 61, 62, 66 → mean = **51.86** (no change)

Centroids did not change → algorithm stops (converged).

Final Clusters:

Cluster	Members	Center
Cluster 1 (Younger)	16–41 (12 values)	23.5
Cluster 2 (Older)	42–66 (7 values)	51.86

Conclusion: The visitors naturally split into a **younger group (around age 23)** and an **older group (around age 52)**.

13. Explain the advantages and disadvantages of Naïve Bayes Classifier.

Naïve Bayes is a classification method based on **Bayes Theorem**. It is called "naïve" because it assumes all features are **independent** of each other (which is rarely fully true, but still works well).

Advantages:

1. **Simple and easy** to build and understand.
2. **Very fast** — works well even on large datasets.
3. **Needs less training data** than many other methods.
4. **Great for text data** — used in spam filtering and sentiment analysis.
5. Handles **multiple classes** easily.

Disadvantages:

1. **Independence assumption** — in real life, features are often related, which can reduce accuracy.
2. **Zero-frequency problem** — if a category was never seen in training, it gives zero probability (fixed using smoothing).
3. **Not ideal for numeric data** with complex relationships.
4. Its probability values are **not always very accurate**, even if the final class is correct.

Conclusion: Naïve Bayes is fast, simple, and excellent for text classification, but its assumption of independent features is its main weakness.

14. Discuss Parameter Tuning and Optimization

Parameter Tuning means **finding the best settings (parameters) of a model** so that it gives the most accurate results. These settings are chosen **before training** and are called **hyperparameters** (e.g., K in K-Means, depth of a decision tree, learning rate).

Why needed: The right parameters can greatly improve a model's accuracy, while wrong ones can make it perform poorly.

Common methods of tuning:

1. **Grid Search** — try **all combinations** of given parameter values and pick the best.
2. **Random Search** — try **random combinations** (faster than grid search).
3. **Cross-Validation** — test the model on different parts of the data to choose stable parameters.

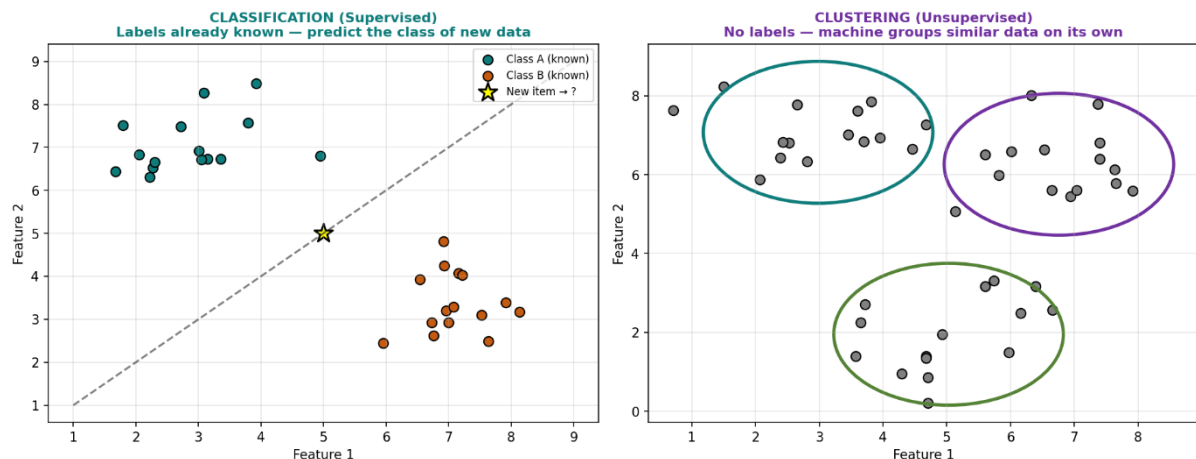
Optimization means **adjusting the model to reduce errors and improve performance** — for example, lowering the difference between predicted and actual values.

Example: For a decision tree, tuning the "maximum depth" prevents the tree from becoming too big (overfitting) or too small (underfitting).

Conclusion: Parameter tuning and optimization help build a model that is accurate, balanced, and works well on new data.

15. Write a short note on Classification.

Classification vs Clustering



Classification is a **supervised** data mining technique used to **predict the category (class) of new data**, based on patterns learned from **labeled training data**.

Key points:

- The training data already has **known labels (classes)**.
- The model **learns** from this data, then predicts the class of **new, unseen data**.
- The output is a **predefined category**.

Steps:

1. **Training** — the model learns from labeled data.
2. **Testing** — the model is checked on new data to measure accuracy.
3. **Prediction** — the model assigns a class to new data.

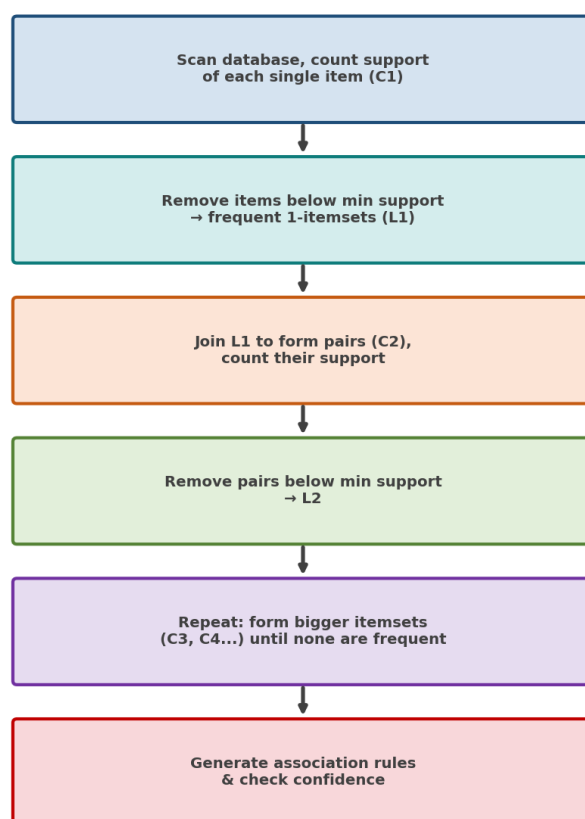
Examples: Email as Spam/Not Spam, loan as Approved/Rejected, tumor as Benign/Malignant.

Common algorithms: Decision Tree, Naïve Bayes, Logistic Regression, KNN.

Conclusion: Classification sorts new data into known groups by learning from past labeled examples.

16. Explain Association Rule and its types.

Apriori Algorithm — Flow



An **Association Rule** shows a **relationship between items** that frequently occur together, written as $A \rightarrow B$ ("if A is bought, B is likely bought too"). It is mainly used in market basket analysis.

Types of Association Rules:

1. **Single-dimensional rule** — involves **one type (dimension)** of item.
 - *Example:* buys(bread) $\rightarrow$ buys(butter).
2. **Multi-dimensional rule** — involves **more than one dimension/attribute**.
 - *Example:* age(young) $\wedge$ income(high) $\rightarrow$ buys(laptop).
3. **Boolean Association Rule** — checks only **whether an item is present or not** (yes/no).
 - *Example:* {milk, bread} $\rightarrow$ {butter}.
4. **Quantitative Association Rule** — involves **numeric values or quantities**.
 - *Example:* age(30–40) $\wedge$ salary(50k–70k) $\rightarrow$ buys(car).

Conclusion: Association rules come in different types depending on the number of dimensions and whether the data is yes/no or numeric — all used to find useful "what-goes-with-what" patterns.